# SymPy Bug Card

## Bug 23: solveset returns 0 for atan(x) = pi

**Task.** Solve the principal inverse-tangent equation over the complex domain.

$$\mathrm{solveset}(\mathrm{Eq}(\mathrm{atan}(x), \pi), x, \mathbb{C})$$

**SymPy's answer.**

$$\mathrm{solveset}(\mathrm{Eq}(\mathrm{atan}(x), \pi), x, \mathbb{C}) = \{0\}.$$

**Correct answer.**

$$\varnothing$$

**Explanation.** Substituting SymPy's returned value gives $\mathrm{atan}(0) - \pi = -\pi$, so 0 is not a solution. The error comes from treating tan as a globally valid inverse of the principal branch atan, even though $\mathrm{atan}(\tan(\pi)) = 0$, not $\pi$.

**Likely root cause.** The diagnosis points to `sympy/solvers/solveset.py:_invert_complex`, where the generic inverse-function branch applies `atan(x).inverse() == tan` without a principal-range condition and later checks only definedness. This is a new instance of a known failure mode.

**Artifact (GitHub).** `bug-report/bug-023-solveset-returns-0-for-atan-x-pi/`